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Data Type

In Python, unlike statically typed languages like C or Java, there is no need to specifically declare the data type of the variable. In dynamically typed languages such as Python, the interpreter itself predicts the data type of the Python Variable based on the type of value assigned to that variable.

Core Data Types in Python

As the name suggests, a data type is the classification of the type of values that can be assigned to variables. We have already seen that, here in Python, we don't need to declare a variable with explicitly mentioning the data type, but it's still important to understand the different types of values that can be assigned to variables in Python. After all the data type of a variable is decided based on the value assigned.

Python data types are categorized into two as follows:

- **Mutable Data Types:** Data types in python where the value assigned to a variable can be changed. Some mutable data types in Python include set, list, user-defined classes and dictionary.
- **Immutable Data Types:** Data types in python where the value assigned to a variable cannot be changed. Some immutable data types in Python are int, decimal, float, tuple, bool, range and string.

Following diagram lists the data types that fall under the categories of the mutable and immutable data types in python:



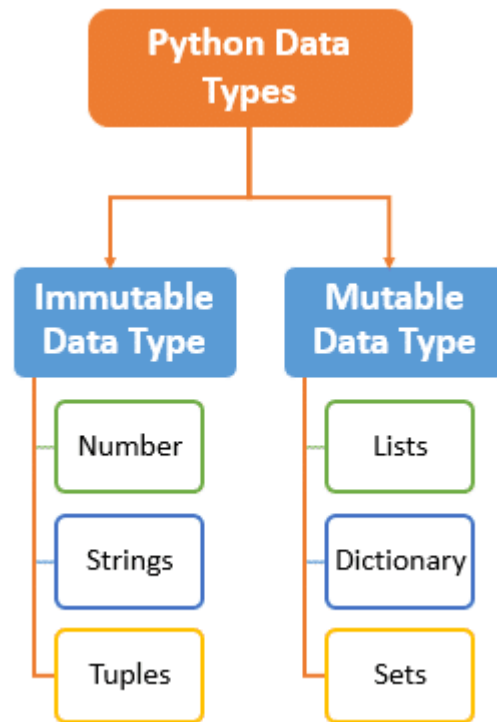
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Let's discuss the above-mentioned core data types in Python.

- **Numbers:** The number data type in Python is used to store numerical values. It is used to carry out normal mathematical operations.
- **Strings:** Strings in Python are used to store textual information. They are used to carry out operations that perform positional ordering among items.
- **Lists:** The list data type is the most generic Python data type. Lists can consist of a collection of mixed data types, stored by relative positions.
- **Tuples:** Python Tuples are one among the immutable Python data types that can store values of mixed data types. They are basically a list that cannot be changed.
- **Sets:** Sets in Python are a data type that can be considered as an unordered collection of data without any duplicate items.
- **Dictionaries:** Dictionaries in Python can store multiple objects, but unlike lists, in dictionaries, the objects are stored by keys and not by positions.



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Built-in Data Types in Python

The categories of built-in data types in Python are:

- **Binary Types** – bytes, memory view, bytearray
- **Mapping Type** – dict
- **Numeric Type** – int, float, complex
- **Text Type** – str
- **Boolean Type** – bool
- **Set Types** – set, frozenset
- **Sequence Types** – list, range, tuple

How to check Data Type in Python

Now that you know about the built-in data types and the core data types in Python, let's see how to check data types in Python.

You can get the data type of any object by using the `type()` function.



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